

Deployment Recommendation Report

Customer-Support Assistant for NorthLink Telecom: where it should run and which cloud services it should use

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Use case title	LLM customer-support chat assistant for a Canadian telecom (from the sample list: "Customer-support assistant for a telecom")
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GitHub commit link	[PASTE COMMIT LINK HERE] (file: docs/assignment1/INFT41000_A1_Janu_Divyansh.pdf)

AI-use statement

AI tool used: Claude (Anthropic), October 2026.

Purpose: The AI was only used for the following three purposes allowed by the assignment guidelines: (a) understanding the course materials that I had missed in class, (b) proofreading for any grammar and spelling errors, and (c) organizing the notes into the form of a report through brainstorming. I have not used AI for scoring, evidence, analysis, recommendations, costing, services, diagrams, or references.

Note: NorthLink Telecom is an imaginary company, one that I invented for the purposes of this task. The details about it (customers, team numbers, systems, chats, etc.) are my own assumption.

1. Use case and readiness snapshot

1.1 Use case

NorthLink Telecom is a mid-sized telecom company operating in Canada that has 2 million mobile and internet customers in Ontario and Quebec provinces. This organization needs an assistant chatbot to operate on its website and app. Users: Validated users (English-speaking or French-speaking), and human representatives managing conversations that are out of the bot's capabilities. Functions: Offer help with billing inquiries such as "Why is my bill so high this month?", provide guidance on how to solve the most frequent issues, identify situations when a transfer to a human agent is required. Data sources: Approximately 1,200 articles related to assistance and plans information about customer's account and billing history (read-only access, coming from the existing on-premises billing system) and the conversation itself. Model: Hosted LLM that provides answers to questions based on the articles (retrieval augmented generation (RAG)) and small classifier that detects which conversations need human attention. NorthLink does not train the LLM. The lifecycle stage of critical importance: Post-launch monitoring. Plans and prices are updated every month. thus, the knowledge that was valid at the moment of launch might become outdated (Huyen, 2022, ch. 8).

1.2 Why it has to be production AI

This system would serve up to two million customers at any hour, and it gets busiest during outages and on bill day, which is exactly when the human queue is longest. A wrong answer about a bill or a refund is not just a bug in a demo. A tribunal made the airline cover for the claims of the chatbot of their airline, making Northlink liable for all responses of its chatbot in *Moffatt v. Air Canada* (2024). It is also a processor of personal data and thus needs uptime, security, and privacy from day one.

1.3 Readiness Scorecard (full scorecard in Appendix A)

Dimension	Score	Evidence	Gap type
Data	3	Review of the 1,200 help documents takes place quarterly by the content team, but there is no versioning which links the document to the relevant price list.	Monitor after launch
Model	3	Prototype answered correctly 123 out of 150 randomly selected billing questions (82%) in one trial, without using any reuseable test dataset, without performing any French testing, and without drift detection.	Fix before launch
Infrastructure	1	Prototype is in Jupyter Notebook running on developer laptop making calls to publicly exposed LLM API endpoint, no deployment, monitoring, and rollback is possible.	Fix before launch
Organizational	2	The Director of Customer Care is responsible for this on the business side, although there is not any technical owner, nor ML Ops team, nor on call rotations.	Fix before launch
Risk	3	Wrong billing responses, personal information leaks, and injection were mentioned in the risk workshop. However, there has been no PIA, nor any mitigation action so far.	Fix before launch
Cost	3	2.4 estimates the cost to be around CAD 0.75 million for three years, but no one has ever quantified the agent hours that would be saved.	Monitor after launch
Total	15 / 30	Rating: 2.5 on average (out of 5). Weakest areas: Infrastructure (1), Organizational (2). As there is a '1', discussion will be required prior to launch.	

1.4 How the two weakest scores affect the decision

Infrastructure (1). Nothing is deployed yet, so whatever we pick has to be built from scratch. Building our own GPU serving setup on-premise (servers, drivers, a model server, scaling and failover) would take months before a single customer could use it. This pushes me toward **managed and serverless cloud services** that already come with scaling, monitoring and rollback, set up as code with AWS CDK so we can roll back from the first release.

Organizational (2). NorthLink has a strong network operations team, but no one who runs LLMs or ML pipelines, and no on-call rotation for this system. Hosting our own model would need that team around the clock. This pushes me toward **buying** the model as a managed service (Section 3.4) and keeping the part we build ourselves small. It also makes "name a technical owner and set up on-call" a condition for Go (Section 4.2).

2. Deployment environment evaluation

2.1 Environment Fit Scorecard (full scorecard in Appendix B)

Question (in order)	Score	Evidence for this use case
1. Data sensitivity	3	Chats include names, account numbers and billing history, which is personal information under PIPEDA. But the help articles are public, and PIPEDA allows a service provider to process data, even across borders, as long as NorthLink stays accountable through contracts and notice (OPC, 2009).
2. Cost pattern	5	I estimate around 150,000 monthly conversations, but there are 4 times as many during downtime and bill day, so any GPUs we owned would have been largely idle.
3. Operational capacity	4	Although NorthLink has an around-the-clock network operations center, no one works on GPUs, models or ML pipelines there.
4. Latency	4	It does not matter for the customers when the response begins after about 2 seconds, but 20-40 ms of the network trip is negligible compared with the time spent by the LLM on writing the response.

Using the decision criteria in order. (1) Limits: since neither data sensitivity nor latency is a 1, public cloud is not excluded. (2) Distribution: $5 - 3 = 2$, and this is less than 3 which is the hybrid threshold. (3) Average: $(3 + 5 + 4 + 4) / 4 = 4.0$, and this is 3.6 and above which indicates that according to the decision criteria, cloud should be used. The score that needs to be watched carefully is the data sensitivity score of 3.

2.2 Comparing the three environments (full Table B2 in Appendix B)

On-premise gives NorthLink the most control over its data, but it would have to buy GPUs big enough for the worst outage hour and hire a 24/7 team to run them. Cloud fits the up-and-down demand and needs the fewest new people, but the conversation text is processed outside Canada for a few seconds (see 2.3). For this system, a hybrid setup (serve on-premise, train in the cloud) would keep the hardest part, serving during spikes, on our own hardware, so we would pay for both and still not get the scaling. Table B2 spells out what each option means for this system specifically.

2.3 Recommendation

My recommendation: cloud. The complete assistant operates in the AWS Cloud within the Canada (Central) region. The billing system is on-premises, while the assistant accesses it through the private VPN.

The two factors that determined the decision were cost structure and operational capabilities. Demand is unpredictable: an outage can multiply chats by four within an hour. Pay-per-use services like Lambda

and Bedrock cost nothing when they are idle, while servers we owned would have to be sized for the busiest hour. NorthLink also has nobody who can run a model server around the clock, and the readiness scorecard already shows this (Infrastructure 1, Organizational 2). A managed cloud setup takes most of that work off our hands.

I did not ignore **data sensitivity**. Everything that is stored (sessions, transcripts, vectors, logs and keys) stays in ca-central-1. Claude Haiku 4.5 is not available inside the Canada Region, so requests go through the US cross-Region inference profile. The processing there is temporary and stays on AWS's network, and stored data remains in Canada (AWS, 2025). Bedrock Guardrails masks personal data before the prompt leaves Canada, and the privacy notice and PIA have to mention this (Risk 3).

Connecting to the on-premise billing system is just an integration. All the AI work (serving, retrieval and training the classifier) happens in the cloud, so I would not call this a hybrid design. **The decision rule agrees** with my recommendation (average 4.0, no hard limits).

2.4 Cost over three years (details in Table C2)

Option A (my choice): AWS CA Central with serverless and managed services. Option B: On-premises, running an open-weight LLM on GPU servers that NorthLink purchases for their own data centre. Assumptions: 150,000 chats a month multiplied by 6 iterations is 900,000 LLM invocations a month, each invoking requires around 3,500 input tokens (instructions, 5 retrieved passages, and chat history) and 250 output tokens; the number of calls is fixed over 3 years. Prices are AWS list prices in USD as of October 5, 2026. I applied 10% for regional differences (Canada) and converted in at 1.4254 CAD per USD (Bank of Canada, 2026). The cost of one full-time employee is CAD 162,000 a year: median pay of about CAD 62.50 an hour (WorkBC, n.d.) + 25%. In Option A, there is a need for 0.75 FTE. In Option B, there is a need for 2.5 FTE due to the need of having GPUs and a model server on call 24/7. It requires.

Three-year totals (CAD, rough)	A: AWS cloud	B: On-premise
Compute (Lambda, SageMaker AI vs. 4 GPU servers plus power)	20,000	321,000
Storage and data transfer	10,800	35,600
Managed services / licences (Bedrock tokens, Guardrails, Knowledge Bases vs. hardware support)	351,000	82,100
Staff time to run it	364,500	1,215,000
Three-year total	about 746,000	about 1,654,000

The cloud option costs about **45% of the on-premise option**, which works out to about CAD 0.14 per chat. **The most uncertain line is the Bedrock token cost (High)**. It is 47% of Option A, and it depends directly on chats per month, turns per chat and tokens per turn. I have not measured any of those yet, because the prototype only ran on 150 test questions. If people have longer chats or twice as many customers use it, this line doubles. Prompt caching on the fixed instructions could cut input cost by up to 90% on the cached part but I left that saving out on purpose. The second most uncertain line is on-premise staff (Medium).

2.5 Tipping point

My recommendation changes if NorthLink's privacy officer, or a regulator (for example through Quebec's Law 25 assessment for data leaving Quebec), says conversation text must be **processed** in Canada and not just stored there. Then the US inference profile is no longer allowed, and serving would have to move to a model running in Canada, either an open-weight model on Amazon SageMaker AI in ca-central-1 or GPUs on-premise. Data sensitivity would become a 1 (a hard limit), and the answer would become hybrid or on-premise serving. A second trigger would be steady volume above roughly 4

times today's (about 600,000 chats a month), where the token bill starts to match the cost of owning GPUs.

3. Cloud platform and service selection

3.1 Platform choice: Amazon Web Services (AWS)

- **Existing tools and skills:** I assume NorthLink's customer portal and app login already run on AWS in ca-central-1. That means the assistant can reuse the same login, the same network link to the data centre, and what the team already knows.
- **Data and AI strengths:** Another way to use Claude Haiku 4.5 is by Amazon Bedrock which provides managed API, Guardrails (data masking, grounding), and Knowledge Base (RAG managed). The Knowledge Base and Titan Text Embeddings V2 both run on ca-central-1 region, meaning that knowledge base and vectors reside in Canada (AWS, n.d.-c).
- **Cost:** Lambda, AWS API Gateway Websockets, and DynamoDB On-Demand (serverless services) are priced by request basis and this is appropriate for spiky traffic due to the outages.
- **Lock-in:** the same model is also sold on Google Cloud Vertex AI and Microsoft Foundry (Anthropic, n.d.), and the vector store is open-source PostgreSQL with pgvector, so the most valuable pieces could move later.

Why not Azure: it is the strongest alternative, since some Azure OpenAI models can run in Canadian regions. But NorthLink has no Azure accounts or skills, and using it would mean running a second cloud with a second landing zone. **Why not Google Cloud:** Vertex AI could do the job, but NorthLink has no presence there, so it would bring the same new-platform cost with nothing extra for this workload.

3.2 Service map (full Table C1 with reasons in Appendix C, shown in Figure 1)

Building block	Exact AWS service	Model
Training compute	AI jobs for Amazon SageMaker model training (ml.g5.xlarge, retraining classifier monthly)	PaaS
Serving / hosting	Serverless AWS Lambda used behind Amazon API Gateway (WebSocket API); Amazon SageMaker AI real-time endpoint for classifier	PaaS
Data storage	Amazon S3; Amazon DynamoDB; Amazon Aurora PostgreSQL-Compatible Engine (Serverless v2 + pgvector)	IaaS (S3) / PaaS
Managed AI / ML	Amazon Bedrock (Claude Haiku 4.5, USA cross-region inference); Bedrock Knowledge bases; Bedrock Guardrails; Amazon Titan Text Embeddings V2	PaaS
Additional	Comprehend (PII redaction), Site-to-Site VPN, Web Application Firewall (WAF), CloudWatch, CloudTrail, Config, Budgets, KMS	SaaS / IaaS / PaaS

Ultimately, all decisions will go back to section 2 in which I employed PaaS services for all areas where NorthLink does not have staff to manage things, all data is stored within ca-central-1 region and IaaS components (S3 and the VPN) are mere building blocks and can be easily replaced.

3.3 Containers or serverless (Week 4 order: traffic, then duration, then portability)

Component	Traffic pattern	How long it runs	Portability need	Decision
Chat orchestrator (each turn)	Very spiky: almost nothing at night, 4x during outages	2 to 10 seconds per turn, well under Lambda's 15-minute limit (AWS, n.d.-e)	Low: thin glue code behind our own interface	Serverless (AWS Lambda)

Component	Traffic pattern	How long it runs	Portability need	Decision
Loading new articles	Rare: weekly or when an article changes	A few minutes, run by Knowledge Bases sync	Low: articles stay in S3 in open formats	Serverless (S3 event, Lambda, Knowledge Bases)
Escalation classifier (serving)	Steady: called on every turn, cold starts not OK	Milliseconds, but always on	High: our own model, packaged as a Docker image	Container (SageMaker AI endpoint)
Classifier training	Once a month	Hours on a GPU, longer than Lambda allows	High: same container image	Container (SageMaker AI training job)

The compromise: With serverless computing, we sacrifice control over the runtime environment and incur the inconvenience of cold starts, but we pay zero while idle and never need to worry about capacity planning, which works out for us as we lack ML ops. Containers incur costs even when idle but maintain our single model's portability.

3.4 Build or buy the model

	Buy: Amazon Bedrock (Claude Haiku 4.5)	Build: run an open-weight LLM ourselves (SageMaker AI or EC2 GPUs with vLLM)
Time to value	Weeks: API call, followed by setting up Knowledge Bases and Guardrails	Months: size the GPUs, set up the model server, scaling, safety filters and load tests
Control and customization	Prompting, RAG, Guardrails; version of model is determined by the provider	Full control over weights, fine-tuning and where processing happens (could stay in Canada)
Cost over time	CAD \$0.35 million over 3 years, each for one token, hence very cheap when used little and more expensive when usage increases	Big fixed GPU cost (always on, sized for peaks); only cheaper per token at high, steady volume
Expertise needed	A cloud developer who is proficient in prompts and testing (0.75 person)	ML platform engineers on 24/7 GPU call (about 2.5 people), which NorthLink does not have

Decision: **buy** the LLM through Bedrock, because it is much faster to get running and because of the Organizational score of 2. **Build** only the small escalation classifier, since it captures NorthLink's own rules for when to escalate and is cheap to train. Building the LLM ourselves stays as the backup plan if the tipping point in 2.5 happens.

3.5 Cloud Adoption Framework

Plan. Goal alignment: the bot manages 25% of conversations without an agent, and the customer satisfaction rate is no less than in case of human chat. Pilot scope agreement: web chat, English and French languages, conversations related to billing and plans only. People: assign a technical owner, one cloud / ML engineer and two people from the Support Operations team to undergo training in the basics of AWS. Guarantee funding of \$9,000 CAD per month.

Ready. Before beginning any construction work, create a landing zone with AWS Organizations and AWS Control Tower, with accounts for dev, test, and prod, plus log archive and security accounts. Configure IAM Identity Center with least privilege roles, such that developers have no permission to access transcripts from production. Create a Service Control Policy such that resources are kept in ca-central-1 and Bedrock US Inference Profile is accessed only outside ca-central-1. Create an organization-level CloudTrail trail and KMS key. Notifications on AWS Budgets to be created for 50%, 80%, and 100% of monthly budget with cost tag (app=support-assistant), along with incorporating the Site-to-Site VPN in the billing process using a read-only service account.

Adopt. Integrate the development process with AWS CDK (code as infrastructure), so that every deployment can be rolled back. Import the articles in Knowledge Bases and perform a 500-question test in the test environment. Pilot test in 5% of web chat in four weeks, always showing the “talk to an agent” button. Proceed to 25% and 100% only if the go criteria from Section 4.2 are fulfilled. The billing system is not migrated.

Govern and Operate. The AWS Config rules, and AWS Security Hub will detect publicly exposed S3 buckets, data encryption status, and IAM policy width. CloudWatch dashboards and alarms will monitor response time, error rate, escalation rate, Guardrails blocks, and token spend. The test suite will be re-run every month and whenever there is any change in the article, pricing, and version of the models, with fixed model IDs. Cost is analyzed monthly in Cost Explorer, while the on-call runbook and PIA will be reviewed quarterly.

3.6 Lock-in

Our strongest lock-ins are via the Bedrock Knowledge Base and Bedrock Guardrails due to the absence of a similar configuration, API and grounding check on any other cloud provider. Lambda and API Gateway WebSocket events, together with DynamoDB sessions, take second place. There is less lock-in in relation to the model itself (Claude is sold on three clouds), vectors (pgvector on PostgreSQL), documents (S3 in open formats) and classifier (Docker image). Mitigation: the orchestrator communicates with the model and retrieval via a light-weight internal "LLM gateway" API. Prompts, the test set and the Guardrails rules are stored in Git as plaintext, and annually we verify that it is possible to migrate by testing the test set on another cloud provider’s endpoint.

4. Risks and decision

4.1 Top three risks

Risk (type)	Likelihood	Impact	Mitigation
R1. Wrong billing or policy answer (technical / legal). Real example: in <i>Moffatt v. Air Canada</i> (2024 BCCRT 149), the airline's chatbot made up a bereavement refund rule and Air Canada had to pay the customer.	Medium	High	Answer only from retrieved articles and show the source link. Use the Guardrails grounding check to block unsupported answers. Dollar amounts come only from the billing system, never from the model. The refund goes directly to a human. Get above 95% on the 500 question English/French exam done before Go.
R2. Cloud misconfiguration results in customer data exposure (shared responsibility security). Example: In 2019, due to the misconfiguration of its firewall at AWS, 100 million records of Capital One were exposed, and OCC fined it USD 80 million (2020).	Medium	High	While AWS takes care of the security of the cloud, NorthLink is in charge of how they configure things on the cloud (AWS, n.d.-m). Thus: one role of least privilege per Lambda; S3 Block Public Access; KMS Encryption; No PII in log file (Guardrails Masking); AWS Config/Security Hub notifications; injection test; penetration test before Go.
R3. Processing in the US is not disclosed or approved (compliance). With US inference, chat text is processed in the US for a few seconds.	Medium	Medium	Complete PIA process (incl. Quebec law 25 transfer impact assessment) prior to Go and update privacy notice accordingly. Personal information must be masked prior to the model invocation. US profile should be used instead of Global, so that all processing occurs in North America. In-Canada backup solution must be ready (tipping point 2.5).

I also looked at cost overruns (for example, a flood of bot traffic during an outage). Section 3.5 covers that with AWS Budgets alerts, WAF rate limits, a cap on tokens per reply, and Lambda concurrency limits.

The environment and platform choices hold up, but the readiness scorecard has a 1 (Infrastructure), a 2 (Organizational) and four "fix before launch" gaps. Launching now would put an untested system with no technical owner in front of two million customers. I am not saying No-Go, because the cost and the fit both look good. **We can Go once all of these are true** (target: 10 weeks; owner: the technical owner, reporting to the Director of Customer Care):

- ## 5. Communication and professional practice

AWS Canada (Central) - ca-central-1 — NorthLink AWS production account

All data at rest stays here: knowledge base, sessions, transcripts, logs, keys

Customers
Web chat + mobile app (logged in, EN/FR)

AWS WAF
rate limits, bot rules

Amazon API Gateway
WebSocket API (streams replies)

AWS Lambda
chat orchestrator

Amazon Bedrock Guardrails
PII mask, grounding

Amazon Bedrock
Claude Haiku 4.5
US cross-Region inference profile

Amazon DynamoDB
session state, 30-day TTL

Amazon SageMaker AI
real-time endpoint (escalation classifier)

Amazon Bedrock Knowledge Bases

Amazon S3
help articles + plans

Amazon Titan Text Embeddings V2
(in-Region)

Amazon Aurora
PostgreSQL Serverless v2 + pgvector

Amazon S3
chat transcripts

Amazon Comprehend
PII redaction (sample)

Amazon SageMaker AI
training job (monthly, ml.g5.xlarge)

AWS Site-to-Site VPN

NorthLink on-premise
Existing data centre (Toronto)

Billing API
system of record (read-only for bot)

Agent desk
human handoff queue

Operate & govern (all accounts)
Amazon CloudWatch · AWS CloudTrail · AWS Config · AWS Budgets · AWS KMS · AWS IAM Identity Center
AWS Organizations + AWS Control Tower (dev / test / prod accounts) · deployed with AWS CDK

Legend

- real-time request / response
- .-.-> leaves Canada for transient inference (PII masked)
- .-.- batch / offline flow
- AI / ML service
- compute / integration
- storage / data
- security / network / ops
- environment boundary

1 = customer opens chat; every customer data store sits inside the blue ca-central-1 boundary; only the masked prompt crosses to US inference.

Figure 1. How the NorthLink support assistant is deployed on AWS. All stored customer data is inside the ca-central-1 boundary. Only the masked prompt crosses to US inference (dashed red line). The billing system and agent desk stay on-premise behind the Site-to-Site VPN. Service names match Tables 3.2 and C1.

5.2 Version control and AI use

The final PDF is committed to my course GitHub repository under docs/assignment1/, and the commit link is on the cover page. I did not commit any passwords, keys or personal data (all customer details in this report are made up). My AI-use statement is on the cover page.

References

- Amazon Web Services. (2024, December 6). *Amazon Aurora now available as a quick create vector store in Amazon Bedrock Knowledge Bases*. <https://aws.amazon.com/about-aws/whats-new/2024/12/amazon-aurora-quick-create-vector-store-bedrock-knowledge-bases>
- Amazon Web Services. (2025, November 24). *Accelerate generative AI innovation in Canada with Amazon Bedrock cross-Region inference*. AWS Machine Learning Blog. <https://aws.amazon.com/blogs/machine-learning/accelerate-generative-ai-innovation-in-canada-with-amazon-bedrock-cross-region-inference>
- Amazon Web Services. (n.d.-a). *Amazon Bedrock pricing*. Retrieved October 5, 2026, from <https://aws.amazon.com/bedrock/pricing/>
- Amazon Web Services. (n.d.-b). *Claude Haiku 4.5* [Model card]. Amazon Bedrock User Guide. Retrieved October 5, 2026, from <https://docs.aws.amazon.com/bedrock/latest/userguide/model-card-anthropic-claude-haiku-4-5.html>
- Amazon Web Services. (n.d.-c). *Supported models and Regions for Amazon Bedrock knowledge bases*. Retrieved October 5, 2026, from <https://docs.aws.amazon.com/bedrock/latest/userguide/knowledge-base-supported.html>
- Amazon Web Services. (n.d.-d). *AWS Lambda pricing*. Retrieved October 5, 2026, from <https://aws.amazon.com/lambda/pricing/>
- Amazon Web Services. (n.d.-e). *Lambda quotas*. AWS Lambda Developer Guide. Retrieved October 5, 2026, from <https://docs.aws.amazon.com/lambda/latest/dg/gettingstarted-limits.html>
- Amazon Web Services. (n.d.-f). *Amazon API Gateway pricing* [WebSocket APIs, Canada (Central)]. Retrieved October 9, 2026, from <https://aws.amazon.com/api-gateway/pricing/>
- Amazon Web Services. (n.d.-g). *Amazon Aurora pricing* [Aurora PostgreSQL Serverless v2, Canada (Central)]. Retrieved October 9, 2026, from <https://aws.amazon.com/rds/aurora/pricing/>
- Amazon Web Services. (n.d.-h). *Amazon Comprehend pricing* [Detect PII]. Retrieved October 9, 2026, from <https://aws.amazon.com/comprehend/pricing/>
- Amazon Web Services. (n.d.-i). *Amazon DynamoDB pricing for on-demand capacity* [Canada (Central)]. Retrieved October 9, 2026, from <https://aws.amazon.com/dynamodb/pricing/on-demand/>
- Amazon Web Services. (n.d.-j). *Amazon S3 pricing* [S3 Standard, Canada (Central)]. Retrieved October 9, 2026, from <https://aws.amazon.com/s3/pricing/>
- Amazon Web Services. (n.d.-k). *Amazon SageMaker AI pricing* [On-demand, Canada (Central)]. Retrieved October 9, 2026, from <https://aws.amazon.com/sagemaker/ai/pricing/>
- Amazon Web Services. (n.d.-l). *AWS VPN pricing* [Site-to-Site VPN]. Retrieved October 9, 2026, from <https://aws.amazon.com/vpn/pricing/>
- Amazon Web Services. (n.d.-m). *Shared responsibility model*. Retrieved October 5, 2026, from <https://aws.amazon.com/compliance/shared-responsibility-model/>
- Anthropic. (n.d.). *Claude Haiku 4.5*. Retrieved October 5, 2026, from <https://www.anthropic.com/claude/haiku>
- Bank of Canada. (2026). *Daily exchange rates* [USD/CAD, October 5, 2026]. <https://www.bankofcanada.ca/rates/exchange/daily-exchange-rates/>
- Huyen, C. (2022). *Designing machine learning systems: An iterative process for production-ready applications*. O'Reilly Media.

Mell, P., & Grance, T. (2011). *The NIST definition of cloud computing* (NIST Special Publication 800-145). National Institute of Standards and Technology. <https://doi.org/10.6028/NIST.SP.800-145>

Microsoft. (n.d.). *Microsoft Cloud Adoption Framework for Azure*. Retrieved October 5, 2026, from <https://learn.microsoft.com/en-us/azure/cloud-adoption-framework/>

Modal. (2025). *How much does it cost to run NVIDIA L40S GPUs in 2025?* Retrieved October 5, 2026, from <https://modal.com/blog/nvidia-l40s-price-article>

Moffatt v. Air Canada, 2024 BCCRT 149 (CanLII).

Office of the Comptroller of the Currency. (2020, August 6). *OCC assesses \$80 million civil money penalty against Capital One* [News release 2020-101]. <https://www.occ.gov/news-issuances/news-releases/2020/nr-occ-2020-101.html>

Office of the Privacy Commissioner of Canada. (2009). *Guidelines for processing personal data across borders*. https://www.priv.gc.ca/en/privacy-topics/airports-and-borders/gl_dab_090127/

WorkBC. (n.d.). *Software engineers and designers (NOC 21231)*. Retrieved October 5, 2026, from <https://www.workbc.ca/career-profiles/software-engineers-and-designers>

Appendix A: Completed Readiness Scorecard

INFT 41000 • WEEK 2 • AI USE-CASE READINESS

Production Readiness Scorecard

USE CASE

NorthLink support assistant

PREPARED BY

Divyansh Janu

DATE

10/05/2026

1. Data

Is our data trustworthy enough to run a business decision on?

- 1 No validation, unknown lineage
- 3 Validated, but lineage undocumented
- 5 Validated, versioned, governed

0 1 2 3 4 5

EVIDENCE, ONE SENTENCE

1,200 help articles are reviewed quarterly, but no version history links each article to the price list it describes.

- ☐ Fix before launch ☒ Monitor after launch
☐ No gap

2. Model

Does the model perform reliably outside training conditions?

- 1 Never tested outside training data
- 3 Tested once, no bias/drift plan
- 5 Tested, explainable, monitored for drift

0 1 2 3 4 5

EVIDENCE, ONE SENTENCE

Prototype answered 123 of 150 billing questions (82%) correctly in one test; no repeatable eval set, no French test, no drift plan.

- ☒ Fix before launch ☐ Monitor after launch
☐ No gap

3. Infrastructure

Can our systems handle real load, failure, and change?

- 1 Runs on someone's laptop
- 3 Deployed, but no monitoring/rollback
- 5 Containerized, monitored, autoscaled, recoverable

0 1 2 3 4 5

EVIDENCE, ONE SENTENCE

Prototype is a Jupyter notebook on one laptop calling a public LLM API; no deployment, monitoring or rollback.

- ☒ Fix before launch ☐ Monitor after launch
☐ No gap

READINESS RADAR



TOTAL 15 / 30 AVERAGE 2.5 / 5 LOWEST Infrastructure

GO / NO-GO

Red flag: a score of 0 or 1 needs a leadership conversation before launch.

- ☐ Ship (go) ☒ Delay for remediation
☐ Kill (no-go)

CONDITIONS AND NEXT ACTIONS (OWNER AND DATE)

Go when: technical owner + 24/7 on-call (3 people) named; CDK deploy with tested rollback <15 min and alarms live; >=95% on 500-question EN/FR test set; 0 PII leaks in 300 red-team prompts and PIA signed; 5% pilot for 4 weeks meets CSAT baseline. Owner: technical owner, target 10 weeks.

REVIEWED BY (NAME, ROLE, DATE)

D. Janu (author), 2026-10-05

© Copyright Omar Altair - INFT 41000 Readiness Scorecard. A ready-to-ship use case does not need a 5 everywhere, but a 1 anywhere is a conversation leadership must have first.

Completed course Readiness Scorecard (interactive version), NorthLink support assistant.

Dimension	Score	Evidence	Effect on the deployment decision
Data	3	Content team reviews 1,200 support articles quarterly; there is no version history linking the article with its pricing list.	Archive the articles in the versioned S3 and synchronize Knowledge bases each time; run the test set each time prices change.
Model	3	The prototype managed to answer 123 out of 150 billing questions correctly (82%) only once without using any test set and without having any drift management strategy.	Use a managed model with grounding checks; 500 question test set in two languages turns into a Go situation.
Infrastructure	1	The prototype is a notebook on one laptop invoking a public LLM API, without any deployment, monitoring or rollback.	Ruling out creation of our own GPU infrastructure; opting for serverless deployment on the cloud using AWS CDK as there is rollback facility.
Organizational	2	A business owner is present but there is no technical owner, no ML Ops engineers, and no one-on-call.	Trends towards purchasing (Bedrock) as opposed to self hosting; technical owner and on-call become Go conditions.
Risk	3	Risks have been pointed out (wrong answers, data leaks, prompt injections), but there is still no PIA, and no mitigation either.	Includes Guardrails anonymization, the keeping data in Canada approach, and signing a PIA prior to Go.
Cost	3	The cost estimate is approximately CAD 0.75 million in 3 years, while savings on agent hours have not been quantified.	Benefits are paid on per use; the pilot needs to calculate the number of chat sessions resolved by the bot.

Appendix B: Environment Fit Scorecard and comparison

INFT 41000 • WEEK 3 • AI DEPLOYMENT ENVIRONMENTS

Environment Fit Scorecard

USE CASE

NorthLink support assistant

PREPARED BY

Divyansh Janu

DATE

10/05/2026

1. Data sensitivity

Can this workload's data leave our own network?

1 Regulated or highly sensitive; cannot leave our network

3 Mixed: some regulated data; some not

5 Low sensitivity or de-identified; provider certifications suffice

1

2

3

4

5

EVIDENCE, ONE SENTENCE

Chats hold personal info (names, accounts, billing) under PIPEDA, but articles are public and provider processing is allowed with accountability.

2. Cost pattern

Is demand steady enough to justify owning hardware, or bursty enough to rent it?

1 Steady, predictable, high volume; owning hardware pays off

3 Steady baseline with occasional peaks

5 Unpredictable or bursty; pay-as-you-go beats idle hardware

1

2

3

4

5

EVIDENCE, ONE SENTENCE

About 150,000 chats/month with ~4x spikes during outages and on bill day; owned GPUs would sit idle between peaks.

3. Operational capacity

Do we have the in-house team to run this 24/7, and do we want to?

1 Strong in-house team can run it around the clock

3 Some skills, gaps in on-call or specialist roles

5 No in-house team; the provider must manage the hardware

1

2

3

4

5

EVIDENCE, ONE SENTENCE

24/7 network operations centre exists, but no one runs GPUs, model servers or ML pipelines.

4. Latency

How much delay can this system tolerate between data and compute?

1 Near-zero tolerance; must run beside the data or on site

3 Moderate tolerance; some network delay is acceptable

5 Delay-tolerant; batch or asynchronous work is fine

1

2

3

4

5

EVIDENCE, ONE SENTENCE

First words of a reply within ~2 s are fine; a 20-40 ms network hop is negligible next to generation time.

FIT PROFILE

Data sensitivity: 3

Latency: 4

Operational capacity: 4

Cost pattern: 5

CENTRE: ON-PREMISE

EDGE: CLOUD

AVERAGE

4.0

/ 5

SPREAD

2

HIGH MINUS LOW

HARD LIMITS

None

RECOMMENDATION

Cloud candidate. Average 4.0: elastic demand and low constraints point to cloud.

☐ On-premise

☒ Cloud

☐ Hybrid

ONE-SENTENCE JUSTIFICATION

Cloud (AWS ca-central-1): bursty demand and no in-house ML ops team favour pay-per-use managed services; all data at rest stays in Canada.

QUESTION THAT DROVE THE DECISION

Cost pattern

Rule, in order: a 1 on data or latency rules out public cloud alone. A spread of 3 or more means hybrid. Otherwise 2.4 or less is on-premise, 3.6 or more is cloud, and in between operational capacity breaks the tie.

© Copyright Omar Altrad - INFT 41000 Environment Fit Scorecard. There is no universally best environment, only the best fit for this workload's data, cost, operations and latency.

Completed course Environment Fit Scorecard (interactive version).

Table B1. Environment Fit Scorecard summary

Question (in order)	Facts about my use case	Score (1 = on-prem, 5 = cloud)
1. Data sensitivity	Personal information (names, account numbers, billing history) in chats; the articles are public; PIPEDA allows provider processing if we stay accountable; no law says the data must stay on our network.	3
2. Cost pattern	Around 150,000 chats per month typically, with peaks of around 4x during downtime and bill day.	5
3. Operational capacity	The company has a 24/7 network operations center but lacks skillset in GPU, model deployment, machine learning pipelines, and anything related to machine learning on call.	4
4. Latency	It is sufficient to initiate the response in about 2 seconds; the 20-40 milliseconds on the network link make no difference.	4

Average: 4.0. Spread (highest minus lowest): 2. Data or latency scored 1? No. Rule says: Cloud. My recommendation: Cloud (matches the rule).

Table B2. Environment comparison for my use case

Dimension	On-premise	Cloud (AWS ca-central-1)	Hybrid (serve on-prem, train in cloud)
Data control	Chats, vectors and the model all stay in NorthLink's Toronto data centre; nothing is processed outside Canada.	All stored data stays in Canada under NorthLink's own KMS keys; chat text is processed in the US for a few seconds (masked).	Live chats stay on-premise; only de-identified transcripts go to the cloud for training.
Cost model	CAD 0.36 million for GPU servers and network costs that were upfront irrespective of the number of chats, be it 10,000 or 600,000 per month.	Cost per token and request: roughly \$0.14 per chat, almost nothing at night, more during outages months.	Still spends on GPUs sized for peak usage on-premise, along with cloud cost for training, so it has the highest fixed cost.

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Dimension	On-premise	Cloud (AWS ca-central-1)	Hybrid (serve on-prem, train in cloud)
Scalability	Limited to 4 servers; a 4x outage spike would make customers wait unless we buy for the peak.	Lambda and Bedrock scale with the spike, within account limits (to be raised before Go).	Serving, the part that spikes, is still limited by on-premise hardware.
Operational burden	NorthLink needs to operate GPUs, the driver, the model server, failover, and safety filters all day round (roughly 2.5 persons that NorthLink lacks).	Hardware and model is done by AWS, whereas the settings, test, and on-call (approximately 0.75 headcount) is done by NorthLink.	Both: cloud competencies along with the on-premises serving team.
Best fit for my system?	Only if processing in Canada becomes mandatory (tipping point).	Yes, recommended. It fits the spiky demand and the small team.	No. It keeps the spiky part on fixed hardware; it only makes sense if the tipping point happens.

Appendix C: Service map and cost

Table C1. Service map (one platform: Amazon Web Services, Region ca-central-1)

Building block	Named service	Model	Container / serverless	Why
Training compute	Amazon SageMaker AI training jobs (ml.g5.xlarge)	PaaS	Container	Re-training the escalation classifier once a month (about 4 GPU-hours). Machines are run by AWS, we supply the container image.
Serving / hosting	AWS Lambda + Amazon API Gateway (WebSocket API)	PaaS	Serverless	Logically per each chat turn. streams the answer, dynamically scales on basis of outage peaks free during idle periods.
Serving (classifier)	Amazon SageMaker AI real-time inference endpoint (2 x ml.m5.large)	PaaS	Container	It is always warm, so it always takes quick decisions in each move, 2 examples for availability.
Data storage	Amazon S3 (articles, transcripts)	IaaS	n/a	Versioned storage along with Block Public Access, open formats assist in ensuring that it is easier to move.
Data storage	Amazon DynamoDB (on-demand)	PaaS	Serverless	Session state, 30 days TTL, pay-as-you-go pricing works well for spikes.
Data storage (vectors)	Amazon Aurora PostgreSQL-Compatible Edition, Aurora Serverless v2 with pgvector	PaaS	Serverless	Vector database storage in Canada supported by Knowledge Bases; open-source pgvector.
Managed AI / ML	Amazon Bedrock: Anthropic Claude Haiku 4.5 through the US cross-Region inference profile	PaaS	Serverless	Faster and inexpensive model purchased as an API, without the use of GPUs (the purchase choice).
Managed AI / ML	Amazon Bedrock Knowledge Bases + Amazon Titan Text Embeddings V2	PaaS	Serverless	Managed RAG: splits, embeds (inside Canada) and retrieves articles, with source links.
Additional: safety	Amazon Bedrock Guardrails	PaaS	Serverless	Masks personal data, filters content and checks grounding on both input and output.
Additional: privacy	Amazon Comprehend (PII redaction)	SaaS	Serverless	Design for AI Service that will de-identify information from 10% of a sample of transcripts before training.
Additional: network	AWS Site-to-Site VPN; AWS WAF	IaaS / PaaS	n/a	Secure connection to the billing system through encryption; rate limiting at the boundary.
Additional: ops / governance	Amazon CloudWatch, AWS CloudTrail, AWS Config, AWS Budgets, AWS KMS, IAM Identity Center, AWS Organizations, AWS Control Tower, AWS CDK	PaaS	n/a	Monitoring, audit logs, compliance tests, cost alerts, encryption keys, access control, landing zone, and infrastructure as code.

Table C2. Three-year cost comparison (rough, CAD)

Common Assumptions: Conversations of 150,000 * 6 turns per conversation = 900,000 LLM calls per month; 3,500 input and 250 output tokens per call; volume is constant for 36 months. AWS Costs: on demand pricing, listed price in USD (costs calculated from <https://aws.amazon.com>; October 5 & 9, 2026). If price for US only was available, then 10% additional was included to account for the Canada region (in line with all prices for the Canada (Central) region that were verified), times 1.4254 CAD/USD (Bank of Canada, October 5, 2026). Personnel Cost: CAD 162,000 per year per person (WorkBC, CAD 62.50/hour * 2,080 hours + 25%).

Cost line	Assumption and source (with date)	Option A: AWS cloud	Option B: On-premise	Uncertainty (H/M/L)
Compute	A: Lambda: 1.8 million calls per month x 6 seconds x 1 GB @ 0.0000166667 dollars per GB-second + 0.20 dollars per million requests = approx. 180 dollars per month (AWS, n.d.-d). SageMaker AI in Canada (Central): 2 ml.m5.large endpoint @ 0.128 dollars per hour + 4 hours ml.g5.xlarge training @ 1.56 dollars per hour = approx. 193 dollars per month (AWS, n.d.-k). B: 4 servers with 4 NVIDIA L40S GPUs per server (2 for peak production times, 1 for failover, 1 for development/testing) for \$48,000 per server (Modal, 2025) = \$273,700 plus power and cooling: 4 x 2 kW x 1.5 PUE x 0.15 CAD/kWh x 3 years = \$47,300 (my assumption about facilities cost).	20,000	321,000	M
Storage and data transfer	A: Aurora Serverless v2 – average usage of around 1 ACU @ USD 0.14/ACU-hr (around USD 105/month; AWS, n.d.-g), DynamoDB – about USD 4 (AWS, n.d.-i), S3 200 GB @ USD 0.025/GB - about USD 10 (AWS, n.d.-j), Site-to-Site VPN – 2 connections x USD 0.05/hr, around USD 73 (AWS, n.d.-l), internet egress – around USD 5. B: storage, switches and firewall for the GPU cluster about USD 25,000 (my estimation).	10,800	35,600	L
Managed services / licences	A: Claude Haiku 4.5 at \$1.00 / \$5.00 per million input / output tokens (Anthropic, n.d.) with a 10% regional-profile premium: 3.15 billion input and 225 million output tokens per month, approximately \$4,700. KBs available at \$1.00 per 1,000 requests, approximately \$900. Guardrails (PII + content filters at USD 0.10 + 0.15 per 1,000 text units, 2 units per turn), around USD 450 (AWS, n.d.-a). Embeddings, Comprehend PII sample at USD 0.0001 per 100-character text unit (AWS, n.d.-h), API Gateway WebSocket at USD 1.10 per million messages (AWS, n.d.-f), CloudWatch, WAF and Config, around USD 170. B: Hardware cost support at 10% of the hardware cost per year; model server and safety tools (without license cost).	351,000	82,100	H
Staff time to run it	A: 0.75 of a cloud/ML engineer (configuration, testing, on-call percentage). B: 2.5 individuals (ML platform engineer, GPU/SRE on-call rotation, keeping the ML server alive). WorkBC (n.d.), Accessed on October 5, 2026.	364,500	1,215,000	M
Three-year total	Option A represents about 45% of Option B. The least certain line is the Bedrock token quantity (H), which depends on the number of chats, turns, and tokens, none of which have been quantified yet.	about 746,000	about 1,654,000	